

TOULON ILE DU LEVANT, France

WMO: 076780

Lat: 43.023N

Lon: 6.460E

Elev: 430

StdP: 14.47

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	38.0	40.8	18.9	14.4	42.6	22.4	17.1	44.4	51.2	48.1	46.1	49.6	17.9	290	0.585

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	11.9	86.1	71.5	84.0	70.8	82.2	70.6	75.6	81.3	74.4	80.1	73.2	78.9	11.4	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.8	128.1	79.0	72.6	122.7	78.0	71.3	117.3	77.0	39.4	81.9	38.2	79.9	37.1	78.9	79.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
44.9	39.9	35.0	DB	35.5	90.3	4.5	3.8	32.3	93.0	29.6	95.2	27.1	97.4	23.9	100.1
			WB	29.9	77.2	3.2	1.8	27.6	78.5	25.8	79.6	24.0	80.6	21.7	82.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	61.8	50.3	49.8	53.5	57.7	63.2	71.1	75.3	76.0	70.3	64.3	57.2	52.2	(6)
	DBStd	10.05	3.95	4.14	3.73	3.64	3.93	4.40	3.39	3.59	3.70	4.14	4.15	4.52	(7)
	HDD50	137	44	47	12	0	0	0	0	0	0	0	3	31	(8)
	HDD65	2235	456	426	356	222	79	3	0	0	4	60	234	395	(9)
	CDD50	4456	54	41	122	230	411	634	785	806	610	443	219	101	(10)
	CDD65	1076	0	0	0	1	25	187	320	341	164	38	0	0	(11)
	CDH74	5261	0	0	0	0	51	807	1864	2103	424	12	0	0	(12)
	CDH80	948	0	0	0	0	3	139	331	445	30	0	0	0	(13)
(14) Wind	WSAvg	14.6	16.6	15.7	16.7	15.1	13.9	11.9	12.9	11.7	13.0	14.9	16.3	16.7	(14)
(15) Precipitation	PrecAvg	30.9	3.5	2.4	2.2	2.3	1.8	1.4	0.5	0.7	2.8	3.9	5.1	3.7	(15)
	PrecMax	47.3	10.3	6.7	8.0	5.9	4.6	6.4	2.2	3.9	8.0	8.2	14.6	9.4	(16)
	PrecMin	17.4	0.2	0.0	0.2	0.2	0.1	0.0	0.0	0.1	1.3	1.0	0.5	0.5	(17)
	PrecStd	8.6	2.9	2.0	2.0	1.7	1.3	1.5	0.6	0.9	2.1	1.9	3.5	2.2	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	62.4	61.0	66.8	71.5	79.2	88.1	88.1	90.6	83.2	75.5	67.4	62.7	(19)
		MCWB	53.4	51.6	52.9	58.7	63.5	71.6	70.2	72.5	68.5	65.7	59.7	54.6	(20)
	2%	DB	59.0	58.3	62.9	68.6	74.6	83.8	85.2	86.8	80.4	73.4	65.4	60.3	(21)
		MCWB	51.8	51.0	53.1	58.1	62.9	69.6	71.0	72.2	68.0	64.8	59.4	54.3	(22)
	5%	DB	57.2	56.4	60.7	66.0	71.9	81.1	83.4	84.4	78.2	71.4	63.9	59.0	(23)
		MCWB	52.0	50.5	52.7	57.1	61.8	69.1	70.8	71.4	67.4	63.9	58.8	54.0	(24)
	10%	DB	55.6	55.0	58.9	63.7	69.7	78.8	81.7	82.4	76.4	69.6	62.6	57.7	(25)
		MCWB	51.2	50.1	51.6	55.9	60.9	68.5	70.7	71.4	66.7	63.5	58.0	53.4	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	55.9	54.8	56.6	61.9	67.5	75.3	77.0	77.5	73.7	69.5	63.3	58.4	(27)
		MCDB	59.0	57.7	61.4	67.3	72.8	81.9	83.3	83.4	78.4	72.2	65.3	59.7	(28)
	2%	WB	54.6	53.6	55.3	59.9	65.4	73.4	75.3	76.1	71.7	67.8	61.7	56.6	(29)
		MCDB	57.3	56.0	59.7	65.3	71.4	79.5	80.8	81.9	76.5	70.7	63.7	58.4	(30)
	5%	WB	53.5	52.3	54.3	58.5	63.8	71.9	74.2	74.8	70.1	66.3	60.3	55.5	(31)
		MCDB	56.2	55.1	58.2	63.4	69.6	77.9	79.6	80.5	75.1	69.1	62.5	57.7	(32)
	10%	WB	52.2	51.1	53.3	57.3	62.6	70.2	73.0	73.5	68.7	65.1	59.1	54.4	(33)
		MCDB	55.0	54.2	57.0	62.0	67.8	76.0	78.5	79.2	73.8	68.1	61.5	57.0	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	7.0	8.4	9.6	10.2	11.0	12.0	12.3	11.9	10.6	8.4	7.3	6.8	(35)
		MCDBR	7.6	9.5	11.5	12.7	13.7	14.7	13.9	13.9	12.0	10.0	8.0	6.9	(36)
	5% WB	MCWBR	6.4	7.4	7.0	7.3	8.1	7.4	7.3	7.9	7.3	6.3	6.3	5.8	(37)
		MCWBR	6.5	8.0	9.5	11.2	11.5	12.7	11.8	12.5	10.3	8.3	6.3	6.2	(38)
(40) Clear-Sky Solar Irradiance	taub		0.325	0.342	0.376	0.398	0.406	0.406	0.401	0.397	0.389	0.384	0.351	0.322	(40)
		taud	2.491	2.444	2.366	2.327	2.345	2.379	2.396	2.414	2.422	2.423	2.461	2.503	(41)
	Ebn at Noon		257	271	274	276	275	275	273	273	266	252	244	247	(42)
		Edn at Noon	23	28	34	38	39	38	37	35	32	29	24	21	(43)
(44) All-Sky Solar Radiation	RadAvg		587	867	1312	1698	2031	2346	2347	2048	1544	1000	640	509	(44)
	RadStd		56	88	111	111	151	91	96	71	80	80	64	54	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (25 neighbors)	+0.52	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page